used in this analysis were conducted in the period 1976-1981. Controlling for $PM_{2.5}$, two-week average ozone has highly variable association with RRADs and MRADs. Controlling for ozone, two-week average $PM_{2.5}$ was significantly linked to both health endpoints in most years.

Minor Restricted Activity Days

Using the results of the two-pollutant model, we developed separate coefficients for each year in the analysis, which were then combined for use in this analysis. The coefficient is a weighted average of the coefficients in Ostro and Rothschild (1989, Table 4) using the inverse of the variance as the weight:

$$\beta = \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2}} \right) = 0.00741.$$

The standard error of the coefficient is calculated as follows, assuming that the estimated year-specific coefficients are independent:

$$\sigma_\beta^2 = \mathrm{var}\left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2}} \right) = \mathrm{var}\left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\gamma} \right) = \sum_{i=1976}^{1981} \mathrm{var}\left(\frac{\beta_i}{\sigma_{\beta_i}^2 \times \gamma} \right)$$

This reduces down to:

$$\sigma_\beta^2 = \frac{1}{\gamma} \Rightarrow \sigma_\beta = \sqrt{\frac{1}{\gamma}} = 0.00036.$$

Incidence Rate: daily incidence rate for minor restricted activity days (MRAD) = 0.02137 (Ostro and Rothschild, 1989, p. 243)

Population: adult population ages 18 to 64

E.5.3 Parker et al. (2009)

Parker et al. (2009) investigated the associations between long-term $PM_{2.5}$ exposure and respiratory allergies in an unrestricted population of children (aged 3-17 years) sampled from the United States National Health Interview Survey. Authors obtained symptom data from participant parents, who reported respiratory allergies on annual surveys. Parker et al. (2009) placed all study participants reporting symptoms of respiratory allergies or hay fever into a combined rhinitis group. Parker et al. (2009) then linked annual averages of SO_2, NO_2, $PM_{2.5}$, and $PM_{2.5\text{-}10}$ and warm season (May to September) O_3 averages to participant's addresses through ambient air pollution and meteorological data (O_3, SO_2, NO_2, $PM_{2.5}$, and $PM_{10-2.5}$) collected from US EPA Air Quality System monitors. The authors adjusted their logistic regression models for survey year, poverty-level, race/ethnicity, age, family structure, insurance coverage, usual source of care, education of adult, urban-rural status, region, and median county-level income.

Incidence, Hay Fever/Rhinitis

In a multi-pollutant model, the coefficient and standard error were estimated from an odds ratio of 1.29 (95% CI: 1.07-1.56) for a 10 μg/m^3 increase in $PM_{2.5}$ concentrations (Parker et al. 2009, Table 4).

E.6 Asthma-Related Effects

Table E-6 summarizes the health impacts functions used to estimate the relationship between $PM_{2.5}$ and asthma exacerbation. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-6. Core Health Impact Functions for Particulate Matter and Asthma-Related Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Asthma Symptoms (Albuterol Use), Albuterol use	Rabinovitch et al.	2006	Denver, CO	6-17		D24HourMean	0.001996	0.001477	Log-linear	Albuterol use
Asthma Onset	Tétreault et al.	2016	Québec, Canada	0-17		Annual	0.043672	0.000885	Log-linear	Separate HIFs for ages 0-4; 5-17

E.6.1 Rabinovitch et al. (2006)

Rabinovitch et al. (2006) analyzed the relationship between short-term $PM_{2.5}$ exposure and asthma exacerbation in children. The study followed children, ages 6 to 13 attending the Kunsberg School at the National Jewish Medical Research Center with diagnosed asthma for two consecutive winters from 2001-2003. Authors gave an electronic bronchodilator (albuterol) to the children to capture the frequency of use within a 24-hour period. The children also responded to three questions to determine if they may have an upper respiratory infection (URI), and urine samples were taken to measure urinary leukotriene E_4 levels on select days. The authors collected hourly ambient $PM_{2.5}$ levels from the Colorado Department of Health Air Pollution Control Division's Tapered Element Oscillating Microbalance (TEOM) monitor, located 2.7 miles west of the school. Additionally, a Federal Reference Monitor (FRM) located next to the TEOM measured 24-hour $PM_{2.5}$ levels. The authors obtained meteorological data from the Colorado Department of Health Air Pollution Control Division and the National Climatic Data Center. A Poisson regression modeled albuterol use as a function of the morning (12:00am to 11:00 am) maximum hourly $PM_{2.5}$ level or the morning mean hourly $PM_{2.5}$ level. The model used both the TEOM and FRM data, individually, incorporated four lag periods (0 to 2 days and 0- to 2-day average), and included several covariates: temperature, pressure, humidity, time trend, Friday indicator, and URI indicator. Rabinovitch et al. (2006) found that, although the $PM_{2.5}$ pollution levels were well below the National Ambient Air Quality Standards, there is a consistent association between peak ambient $PM_{2.5}$ levels and increased albuterol use in asthmatic children.

Asthma Symptoms (Albuterol Use), Albuterol Use

In a single-pollutant model, the coefficient and standard error were estimated from a percentage of use increase of 1.2% (95% CI: -0.6-2.9%) for a 6 µg/m^3 increase in averaged daily mean $PM_{2.5}$ concentration lagged by 0-, 1-, and 2-days (Rabinovitch et al. 2006, Table 4, pg. 1099).

E.6.2 Tétreault et al. (2016)

Tétreault et al. (2016) investigated the relationship between childhood asthma onset and long-term pollution exposure ($PM_{2.5}$, NO_2, O_3) in Quebec, Canada. The authors obtained data from four medical-administrative databases collectively known as Quebec Integrated Chronic Disease Surveillance System (QICDSS) between April 1, 1996 and March 31, 2011. The study defined the onset of asthma as a hospital discharged diagnosis of asthma or two reports of asthma from two separate physicians within a two-year period. The authors used Cox proportional hazard models to estimate the association between asthma onset and pollution exposure, controlling for population subgroup status. Time-varying exposure models assessed time-varying exposures to the three pollutants in question. Tétreault et al. (2016) showed that childhood asthma onset may be associated with exposure to $PM_{2.5}$, NO_2, and O_3.

As the physiology and etiology of lung development in children is similar in children 6-17, we apply the 4-12 year age-striated effect estimate from Tétreault et al. (2016) to children ages 4-17 (Baena-Cagnani et al., 2007, Guerra et al., 2004, Ochs et al., 2004, Sparrow et al., 1991, Trivedi and Denton, 2019).

Incidence, Asthma

In a single-pollutant time-varying model, the coefficient and standard error were estimated from a hazard ratio of 1.33 (95% CI: 1.31-1.34) for a 6.53 µg/m^3 (interquartile range) increase in annual $PM_{2.5}$ concentration at the residential address (Tétreault et al. 2016, Table 5).

E.7 Sensitivity Analysis – General

Table E-7 summarizes the $PM_{2.5}$ health impacts functions considered by EPA to be sensitivity analyses. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-7. Core Health Impact Functions for Particulate Matter Sensitivity Analyses

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Di et al.	2017	Nationwide	65-99		Annual	0.008066	0.000118	Log-linear	Single-pollutant model
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O_3	Annual	0.005921	0.000096	Log-linear	Nearest monitor analysis
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O_3	Annual	0.007789	0.000118	Log-linear	Cox model with mixed effects
Hospital Admissions, Respiratory	Jones et al.	2015	New York State	0-99		D24HourMean	0.000800	0.000170	Logistic	HA, Respiratory-1
Incidence, Asthma	McConnell et al.	2010	13 Southern California communities	4-17		Annual	0.029127	0.017732	Log-linear	
Incidence, Asthma	Nishimura et al.	2013	5 Urban regions	7-21		Annual	0.029559	0.069101	Logistic	Black, Hispanic
Mortality, All Cause	Pope et al.	2015	Nationwide	30-99		Annual	0.006766	0.000712	Log-linear	LURBME model
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	Massachusetts	0-99	O_3	D24HourMean	0.000499	0.000355	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	New Jersey	0-99	O_3	D24HourMean	0.001094	0.000227	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	New Mexico	0-99	O_3	D24HourMean	0.001094	0.001943	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	New York	0-99	O_3	D24HourMean	0.001094	0.000151	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	Florida	0-99	O_3	D24HourMean	-0.000401	0.000307	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	New Hampshire	0-99	O_3	D24HourMean	-0.001207	0.001238	Logistic	
Hospital Admissions, All Cardiac Outcomes	Talbott et al.	2014	Washington	0-99	O_3	D24HourMean	-0.000904	0.000540	Logistic	